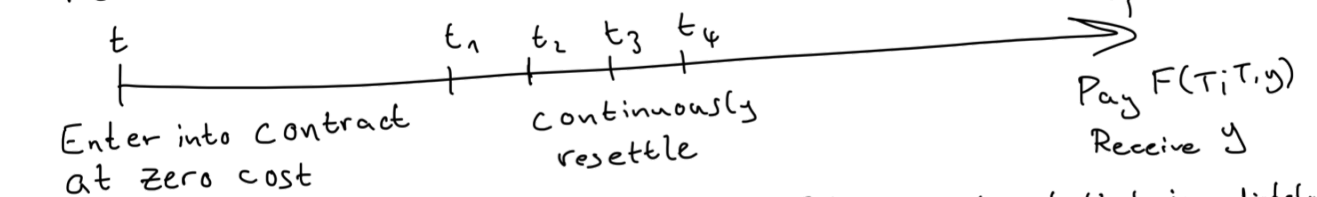


Forward contracts: These are over-the-counter agreements settled at a specific expiration date. Because money only changes hands at maturity, parties are exposed to credit risk (e.g. counterparties may default at maturity).

Futures contract: These are exchange-traded contracts that are continuously resettled daily, which eliminates credit risk. Let $F(t;T,Y)$ be the price of the futures contract. At every $s \leq T$, there is a quoted futures price $F(s;T,Y)$, which makes the futures contract on the claim Y , if entered at s , zero. At expiration T , the holder of the contract pays $F(T;T,Y)$ and receives Y from the underwriter. During any infinitesimal time interval $(s, s+\Delta s)$, the holder of the contract receives or pays the amount $F(s+\Delta s;T,Y) - F(s;T,Y)$.



Note that at time T we can enter into a futures contract that immediately expires. We thus need to pay $F(T;T,Y)$ and receive Y . Absence of arbitrage therefore requires that:

$$F(T,T,Y) = Y$$

Which means that if we can show that futures prices are martingales, then we can conclude:

$$F(t,T,Y) = E_t^Q[Y] \quad (\text{Proof in slides}).$$

Problem with modelling commodities:

Seasonality: Seasonal peaks driven by production constraints or demand cycles.

Convenience yield and storage:

Storing physical commodities is expensive, but holding inventory acts as vital "insurance" against supply disruptions or sudden demand spikes. The implicit benefit of holding the physical asset is called the convenience yield. Convenience yield is inversely related to inventory levels.

Backwardation vs. Contango:

When inventories are low, the convenience yield is high, meaning the spot price is often higher than the futures price. Conversely, high inventories typically result in an upward-sloping curve known as contango.

Combine storage cost and convenience yield into net convenience yield δ .

Assuming constant interest rates and net convenience yield, the futures price would be given by:

$$F(t,T) = S(t) e^{(r-\delta)(T-t)}$$

$$S(t) = r(t) - \frac{\log(F(t,T+\Delta T)) - \log(F(t,T))}{\Delta T}$$

We can combine r and δ into one rate, known as the cost of carry.

Let X_t be an affine factor:

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t^Q$$

$$\mu(X_t) = K_0 + K_1 X_t \quad \sigma(X_t) = H_0 + H_1 X_t$$

$$r_t = \rho_0 + \rho_1 X_t$$

$$P(t,T) = E^Q \left[e^{\int_t^T (\rho_0 + \rho_1 X_u) du} \right] = e^{A(T-t) + B(T-t) X_t}$$

Spot price of a commodity is given by $S_t = e^{\delta_0 + \delta_1 X_t}$. Then we have:

$$F(t,T) = E_t^Q[S_T] = E_t^Q[e^{\delta_0 + \delta_1 X_T}] = e^{A(T-t) + B(T-t) X_t}$$

where $A(\cdot)$ and $B(\cdot)$ solve:

$$\frac{dB(\tau)}{d\tau} = K_1^T B(\tau) + \frac{1}{2} B^T(\tau) H_1 B(\tau), \quad B(0) = \delta_1$$

$$\frac{dA(\tau)}{d\tau} = K_0 B(\tau) + \frac{1}{2} B^T(\tau) H_0 B(\tau), \quad A(0) = \delta_0$$

Schwartz one-factor model:

Assume that the spot price is given by: $S_t = e^{x_t}$, where x_t is an Ornstein-Uhlenbeck process, whose dynamics under the risk-neutral measure are given by:

$$dx_t = \kappa(\theta - x_t)dt + \sigma dW_t^Q$$

$$dS_t = \kappa\left(\theta - \frac{\sigma^2}{2\kappa} - x_t\right)S_t dt + \sigma S_t dW_t^Q$$

Let's derive the formula for futures prices in this model. Remember that the transform we wish to solve is:

$$F(t,T) = E_t^Q[e^{x_T}]$$

which implies that $\delta_0 = 0$ and $\delta_1 = 1$. The Riccati ODEs are given by:

$$\frac{dB(\tau)}{d\tau} = -\kappa B(\tau), \quad B(0) = 1$$

$$\frac{dA(\tau)}{d\tau} = \kappa \theta B(\tau) + \frac{\sigma^2}{2} B(\tau)^2, \quad A(0) = 0$$

This is solved easily:

$$B(\tau) = e^{-\kappa\tau}$$

$$A(\tau) = \kappa \theta \int_0^\tau e^{-\kappa s} ds + \frac{\sigma^2}{2} \int_0^\tau e^{-2\kappa s} ds$$

$$= -\theta \left(e^{-\kappa\tau} - 1 \right) - \frac{\sigma^2}{4\kappa} \left(e^{-2\kappa\tau} - 1 \right)$$

$$= \theta \left(1 - e^{-\kappa\tau} \right) + \frac{\sigma^2}{4\kappa} \left(1 - e^{-2\kappa\tau} \right)$$

Gibson and Schwartz model:

Assume that the spot price is given by $S_t = e^{x_t}$, where x_t and the latent state δ_t have risk-neutral dynamics:

$$dx_t = (r - \delta_t - \frac{\sigma_x^2}{2})dt + \sigma_x dW_{x,t}^Q$$

$$d\delta_t = \kappa(\alpha - \delta_t)dt + \sigma_\delta dW_{\delta,t}^Q$$

with correlated brownian motions such that $\langle W_{\delta,t}^Q, W_{x,t}^Q \rangle = \rho t$.

Core idea: δ is now its own stochastic process with a brownian motion component that is correlated to the commodity price. Why correlation? Imagine a sudden supply shock (e.g. an oil pipeline burst), the spot price S_t immediately spikes. Because physical oil is suddenly scarce, the benefit of having oil right now skyrockets. Therefore, the convenience yield δ_t must also spike.

The affine coefficients are given by:

$$K_0 = \begin{bmatrix} r - \frac{\sigma_x^2}{2} \\ \kappa\alpha \end{bmatrix}, \quad K_1 = \begin{bmatrix} 0 & -1 \\ 0 & -\kappa \end{bmatrix}, \quad H_0 = \begin{bmatrix} \sigma_x^2 & \rho\sigma_x\sigma_\delta \\ \rho\sigma_x\sigma_\delta & \sigma_\delta^2 \end{bmatrix}$$

Remember $F(t,T) = e^{A(T-t) + B(T-t)X_t}$, where $B(\cdot)$ is the solution to:

$$\frac{dB_x(\tau)}{d\tau} = 0, \quad B_x(0) = 1$$

$$\frac{dB_\delta(\tau)}{d\tau} = -B_x(\tau) - \kappa B_\delta(\tau), \quad B_\delta(0) = 0$$

$$\frac{dA(\tau)}{d\tau} = (r - \frac{\sigma_x^2}{2}) B_x(\tau) + \kappa \alpha B_\delta(\tau) + B^T(\tau) H_0 B(\tau)$$

Schwartz and Smith model:

Assume that the spot price S_t is given by $S_t = e^{\delta_t + x_t}$, where the dynamics of the latent states under the risk-neutral measure are given by:

$$d\delta_t = \nu_t^Q dt + \sigma_\delta^Q dW_\delta^Q(t)$$

$$dx_t = \kappa(\theta_x - x_t)dt + \sigma_x dW_x^Q(t)$$

where $\langle W_\delta^Q, W_x^Q \rangle = \rho t$.

Core idea: The main change is in the spot price. The idea here is that short-term deviations can push the spot price away from some equilibrium price, but that these deviations will slowly dissipate (mean-reversion on x). The equilibrium price itself appears to be best described by a random walk.

This model is of course also affine with futures prices: $\log(F(t,T)) = \delta_t + A(T-t) - x_t$.

Turns out Schwartz and Smith is just a relation of Gibson and Schwartz. That means we can define a Matrix A such that:

$$\begin{bmatrix} \delta_t \\ x_t \end{bmatrix} = A \begin{bmatrix} x_t \\ \delta_t \end{bmatrix}$$

Which means that the two models are formally the same, but Schwartz and Smith might be better because the model is more orthogonal (correlation coefficient is closer to 0).

Adding seasonality:

First way:

Assume that the spot price is given as $\delta_t = g(t) e^{\delta_t + x_t}$, where X_t is an affine process and $g(t)$ is a truncated Fourier series:

$$g(t) = \sum_{n=1}^N [\gamma_n^c \cos(2\pi n t) + \gamma_n^s \sin(2\pi n t)]$$

$$m=1 \rightarrow \text{yearly frequency}, \quad m=2 \rightarrow \text{half-yearly frequency}, \dots$$

Then the futures price is:

$$F(t,T) = E_t^Q[S_T] = e^{g(t)} E_t^Q[e^{\delta_T + x_T}]$$

Second way:

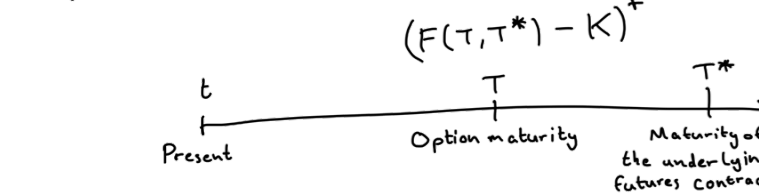
Some commodity prices also exhibit increased/decreased volatility during specific parts of the year. How to model? In a Schwartz-Smith model we could use:

$$\sigma_x(t) = \sigma_x^Q e^{g(t)}$$

where $g(t)$ again models the seasonality. This of course twins the time-inhomogeneity of the affine processes.

Option pricing:

We want to price an option with payoff at time T :



Black 76:

Assume futures dynamics: $\frac{dF_t(t)}{F_t(t)} = \sigma_F dW_t^Q$

Call option price: $C = P(t,T)[F(t,T)\Phi(d_1) - K\Phi(d_2)]$ where Φ is Gaussian cdf with $d_1 = \frac{\log(F(t,T)/K) + \frac{\sigma_F^2}{2}(T-t)}{\sigma_F \sqrt{T-t}}$ and $d_2 = d_1 - \sigma_F \sqrt{T-t}$.

HJM:

Assume futures dynamics: $\frac{dF_t(t)}{F_t(t)} = \sigma_F(t,T) dW_t^Q$

Then we have $X(t,T) = \log(F(t,T))$ with dynamics:

$$dX = \frac{dF}{F} - \frac{1}{2} \left(\frac{dF}{F} \right)^2$$

$$= \sigma_F(t,T) dW_t^Q - \frac{1}{2} [\sigma_F(t,T)]^2 dt + \int_t^T \sigma_F(t,T) dW_u^Q$$

$$X(T,T) = X(t,T) - \frac{1}{2} \int_t^T [\sigma_F(t,T)]^2 du + \int_t^T \sigma_F(t,T) dW_u^Q$$

$$E_t^Q[X(T,T)] = X(t,T) - \frac{1}{2} \int_t^T [\sigma_F(t,T)]^2 du$$

$$Var_t^Q[X(T,T)] = E[(X(T,T) - E[X(T,T)])^2]$$

$$= E[(\int_t^T \sigma_F(t,T) dW_u^Q)^2]$$

$$= \int_t^T \sigma_F(t,T)^2 du$$

Futures can then be written as:

$$F(T,T) = F(t,T) e^{-\frac{1}{2} \int_t^T [\sigma_F(t,T)]^2 du - \int_t^T \sigma_F(t,T) dW_u^Q}, \quad Z \sim N(0,1)$$

The corresponding Call price is then:

$$C(t,T) = P(t,T) E_t^Q[(F(T,T) - K)^+]$$

$$= P(t,T) \int_{-\infty}^{\infty} (F(t,T) e^{-\frac{1}{2} \int_t^T [\sigma_F(t,T)]^2 du - \int_t^T \sigma_F(t,T) dW_u^Q} - K)^+ \phi(z) dz$$

Use Black 76 and just identify volatility:

$$d_1 = \frac{\log(F(t,T)/K) + \frac{1}{2} \int_t^T \sigma_F(t,T)^2 du}{\sqrt{\int_t^T \sigma_F(t,T)^2 du}}$$

Connection to Affine models:

$$F(t,T) = e^{A(T-t) + B(T-t)X_t} \xrightarrow{dF} \frac{dF}{F} = \left\{ (T-t) \sigma(X_t) dW_t^Q \right\}$$

Simply find $\sigma(t,T)^2 = \int_t^T \sigma^2(t,T) du$ and use above Black 76-formula above to find price on option.

Stochastic interest rates and convenience yields:

Define three forward rates:

→ Classic forward rate:

$$P(t,T) = e^{-\int_t^T f(t,u) du}$$

→ Forward convenience yield:

$$F(t,T) = S_t e^{\int_t^T (\delta(t,u) - \delta(t,u)) du}$$

→ Future convenience yield:

$$A(t,T) = S_t e^{\int_t^T (\delta(t,u) - \delta(t,u)) du}$$

Core Idea: Why do Futures and Forwards have different convenience yields? Because $A \neq F$ and the futures price A must incorporate the risk of having to borrow money to cover daily margin calls as interest rates fluctuate.

Dynamics:

$$dS_t = \mu_S(t) S_t dt + \sigma_S(t) S_t dW_t^Q$$

$$d\delta(t,T) = \mu_\delta(t,T) dt + \sigma_\delta(t,T) dW_t^Q$$

We know:

$$\mu_S(t) = r_t - \delta_t$$

$$\mu_\delta(t) = \sigma_F^Q(t,T) \int_t^T \sigma(t,u) du$$

To find μ_δ we simply have to find the dynamics of the exponent in A , use Ito and use the fact that futures prices are Q-martingales i.e. zero drift.

Milner and Schwartz (1998) drift condition:
The drift of the future convenience yield $\mu_\delta(t,T)$ under the equivalent martingale measure is given by

$$\mu_\delta(t,T) = \sigma_\delta(t,T) \left[\int_t^T \sigma_\delta(t,u) du \right] + [\sigma_\delta(t,T) - \sigma_\delta(t,T)] \left[\sigma_\delta(t,T) + \int_t^T (\sigma_\delta(t,s) - \sigma_\delta(t,s)) ds \right]$$

All dynamics:

Q-dynamics in the Milner and Schwartz (1998) model:

$$\frac{dS}{S} = [r(t,t) - \delta(t,t)] dt + \sigma_S(t) dW_t^Q$$

$$d[F(t,T) - \delta(t,T)] = [\sigma_\delta(t,T) - \sigma_\delta(t,T)] dt + \sigma_\delta(t,T) dW_t^Q$$

$$\delta(t,T) = \left\{ \sigma_\delta(t,T) \left[\int_t^T \sigma_\delta(t,s) ds \right] + [\sigma_\delta(t,T) - \sigma_\delta(t,T)] \left[\sigma_\delta(t,T) + \int_t^T (\sigma_\delta(t,s) - \sigma_\delta(t,s)) ds \right] \right\} dt + \sigma_\delta(t,T) dW_t^Q$$

$$\frac{dG(t,T)}{G(t,T)} = \left(\alpha(t) + \int_t^T (\alpha(t,s) - \alpha(t,s)) ds \right) dt + \sigma_G(t,T) dW_t^Q$$

► A model of this type is thus fully determined under Q by specifying the volatility functions σ_S , σ_δ and σ_G .

The Gaussian case:
If we assume that the volatility processes are deterministic, we can find a closed-form solution to the options price in the form of a generalized Black (1976) formula.

Milner and Schwartz (1998) option prices on futures contracts
The time-0 price of a call option with strike price K and maturity $T \geq 0$ on a futures contract $G(t,T)$ is given by

$$C^F = \mathbb{E}^Q \left[e^{-\int_0^T \alpha(u) du} (G(T,T) - K)^+ \right]$$

$$= P(0,T) \left[G(0,T) e^{\alpha(0,T)} \Phi(d_1) - K \Phi(d_2) \right]$$

where

$$d_1 \triangleq \frac{1}{\sigma(t,T)} \left[\log \frac{G(0,T)}{K} + \alpha(t,T) \pm \frac{\nu^2(t,T)}{2} \right]$$

The functions ν and α are given by

$$\nu(t,T) = \int_t^T [\sigma_G(u,T)]^2 du$$

$$\alpha(t,T) = \int_t^T \left[\int_t^u \sigma_G(u,s) ds \right] \sigma_G(u,T) du$$